

Human Resource Management System (HRMS)

Project Overview

Developed a Human Resource Management System (HRMS) using a microservices architecture to automate attendance tracking, leave management, payroll processing, employee authentication, and HR operations. The system integrates with a third-party attendance provider (eTimeOffice) to fetch real-time punch-in and punch-out data, ensuring accurate attendance and salary calculations.

Architecture Summary

- 1 Backend: Spring Boot microservices
- 2 Frontend: React.js
- 3 Security: Spring Security with role-based access control
- 4 Integration: Third-party REST APIs
- 5 Communication: RESTful APIs
- 6 Design: Domain-driven microservices architecture

Microservices Description

1. Attendance Service

Integrated with the eTimeOffice third-party API to fetch employee punch-in and punch-out data. Calculated daily and monthly attendance, total working hours, absent days, and half-days. Attendance data is shared with Leave and Payroll services for further processing.

2. Leave Management Service

Leave service is dependent on attendance data and supports casual, sick, paid, unpaid, and work-from-home leaves. Implemented approval workflow with pending, approved, and rejected statuses. Paid leave is limited to two per month, after which leaves are treated as unpaid. Leave data is reflected in attendance records and leave balance is updated automatically.

3. Employee Service

Handles authentication and authorization with role-based dashboards for employees and admins. Implemented login, logout, and forgot password functionality using OTP-based verification. Acts as the central service for employee information across the system.

4. Payroll Service

Automated payroll calculations based on attendance, working hours, and leave deductions. Generates payslips and offer letters automatically based on salary structure. Offer letters can be downloaded only after employee acceptance and eligibility validation.

5. Employee Mapping Service

Resolved employee ID mismatch between the third-party attendance system and internal employee service by implementing a dedicated mapping service. This ensured consistent and accurate data processing across all microservices.

Key Challenges & Solutions

- 1 Integrated third-party attendance APIs and handled inconsistent employee identifiers.
- 2 Implemented complex business rules for leave and payroll calculations.
- 3 Ensured secure role-based access using Spring Security.

- 4 Automated validations to prevent invalid operations such as early payslip downloads.

Technologies Used

- 1 Java 8
- 2 Spring Boot
- 3 Spring Security
- 4 React.js
- 5 REST APIs
- 6 Third-party API Integration
- 7 Microservices Architecture
- 8 Relational Database
- 9 Git, Postman

Business Impact

The HRMS significantly reduced manual HR effort, improved accuracy in attendance and payroll processing, and provided a scalable and efficient solution for managing employee lifecycle operations.